



Smart City Traffic Optimisation System — Final Documentation

Smart City Traffic Optimisation System (Capstone Documentation)

Project Overview

Urban traffic congestion is a persistent, systems-level problem driven by rapid urbanisation, rising vehicle ownership, constrained road capacity, and imperfect coordination between travellers and traffic-control infrastructure. Congestion causes **nonlinear network effects**: a small disturbance (weather, incident, event surge) can propagate across intersections and corridors, amplifying delay, fuel consumption, and emissions while also reducing reliability for emergency response and public services.

Intelligent Transportation Systems (ITS) address these constraints by shifting traffic management from static, isolated control to **integrated, data-driven decision support** across sensing, prediction, and control. Instead of operating fixed-time plans that assume “average” traffic, ITS leverages real-time and historical data to infer current demand, anticipate near-future states, and recommend (or automate) adaptive actions.

This capstone proposes a **Smart City Traffic Optimisation System** that combines:

- **Historical traffic + weather analysis** (Metro Interstate dataset) to understand temporal and meteorological demand patterns.
- **Analytics dashboards** (Tableau) for operational visibility and KPI tracking.
- **AI-assisted decision support** (LLM-guided rules and modelling rationale) for adaptive signal timing recommendations and emergency prioritisation logic.
- **Mobile-first operator interface** (Figma prototype) for traffic managers and field personnel.
- **Standards-aware documentation** (Notion) to structure requirements, interoperability, and lifecycle considerations.

Motivation: Cities need a practical pathway to modern ITS that is explainable, auditable, and deployable—even with limited engineering resources. A no-/low-code architecture, backed by strong documentation, can reduce implementation friction while enabling measurable mobility and sustainability gains.

Expected societal impact

- Reduced congestion and travel-time variability
- Faster, more predictable emergency response
- Lower fuel consumption and emissions due to reduced idling and stop-and-go traffic
- Improved situational awareness for municipal authorities and traffic police
- A scalable blueprint for broader smart-city deployments

Executive Summary (300–400 words)

This capstone project presents a **Smart City Traffic Optimisation System** designed to support data-driven traffic operations using historical traffic demand, weather context, and AI-assisted decision support. Urban mobility networks are increasingly strained by population growth, rising vehicle volumes, and limited infrastructure expansion. Conventional traffic management—dominated by fixed-time signal plans—often fails under demand volatility and unexpected disruptions, creating recurring congestion, increased fuel use, and unreliable travel times. Intelligent Transportation Systems (ITS) are therefore essential for transforming traffic control into an adaptive, integrated service.

The proposed solution combines three layers: (1) **analytics**, (2) **AI decision support**, and (3) **operator-facing interfaces**. At the analytics layer, the **Metro Interstate Traffic Volume** dataset is used to explore how traffic volume varies by hour, weekday/weekend patterns, seasonal/monthly changes, and weather conditions (rain, snow, cloud cover, temperature). These insights are operationalised in **Tableau dashboards** that allow traffic managers to monitor demand patterns, interpret anomalies, and track performance indicators. At the AI layer, recommendations are generated through a structured workflow: traffic volume and contextual variables → congestion classification → recommended signal timing action (increase/maintain/priority override) → expected operational improvement. This approach is grounded in ITS literature on **adaptive signal control** (e.g., maximum weighted flow concepts) and AI-based prediction/optimisation (including generative AI and reinforcement learning directions).

The project also includes a **Figma-based mobile application prototype** that translates analytics and AI recommendations into a usable operator experience. It documents ten key screens—from authentication to live monitoring, recommendations, emergency prioritisation, notifications, and reporting—ensuring traceability from user needs to interface behaviour. Finally, the system is documented using a standards-aware structure inspired by urban ITS architecture research, ensuring the solution can scale from proof-of-concept to deployment-ready planning.

Overall, the system demonstrates a practical, explainable, and extensible ITS blueprint. It enables better operational decisions, supports emergency service reliability, and promotes sustainability outcomes. Future enhancements include real-time IoT sensors, computer vision, edge AI, connected-vehicle inputs, digital twins, and reinforcement learning for closed-loop traffic control.

Abstract (200–250 words)

Traffic congestion is a critical smart-city challenge that degrades mobility, increases fuel consumption, and elevates emissions. Traditional fixed-time traffic signal control is insufficient under highly dynamic demand and disruption conditions such as adverse weather, incidents, or event-driven surges. This project proposes a **Smart City Traffic Optimisation System** that integrates historical demand analysis, AI-assisted decision support, and operator-focused interfaces to improve traffic management effectiveness.

Using the **Metro Interstate Traffic Volume** dataset, the project performs exploratory and KPI-driven analytics to understand congestion patterns across temporal cycles (hourly, weekly, monthly) and meteorological factors (temperature, rain, snow, cloud cover, and categorical weather states). These insights are operationalised through **Tableau dashboards** to support situational awareness and decision-making. Building on ITS literature regarding adaptive traffic signal control and AI approaches for prediction and optimisation, the system defines a transparent recommendation pipeline: observed traffic volume and context → congestion assessment → recommended signal timing action (increase/maintain/priority override) → expected improvement.

A **mobile dashboard prototype** designed in Figma presents live monitoring, AI recommendations, emergency prioritisation, notifications, and reporting workflows. The overall solution is documented in Notion as a standards-aware, implementation-ready case study. Results indicate that combining analytics with explainable AI recommendations can strengthen operational responsiveness, support emergency reliability, and reduce congestion impacts. The system is designed to be extensible toward real-time sensing, edge AI, connected vehicles, and digital-twin simulations.

Problem Statement

Current urban traffic management faces four persistent issues:

1. **Demand volatility:** traffic fluctuates hourly and seasonally; fixed signal plans assume stable averages.
2. **External disruptions:** weather, holidays, incidents, and special events change driving behaviour and road capacity.
3. **Intersection coupling:** poor coordination across adjacent intersections propagates queue spillback and network-wide delay.
4. **Operational visibility gaps:** decision-makers often lack unified, interpretable insights linking conditions to recommended actions.

Limitations of fixed-time signals

- Optimised for historical averages rather than current demand
- Poor performance under peaks, abnormal weather, or incidents
- Limited ability to prioritise emergency corridors dynamically

- Requires frequent manual retiming cycles, which is costly and slow

Need for adaptive AI-assisted decision support

An adaptive approach is needed to infer congestion risk from multi-factor data and recommend timely control actions with traceable reasoning. AI enables:

- Pattern recognition across large historical datasets
 - Near-term prediction and scenario reasoning
 - Structured recommendations that can be reviewed and audited
-

Objectives (SMART: 8–10)

1. **Develop** a structured documentation framework in Notion covering overview, scope, architecture, dashboards, and UX by **project submission date**.
 2. **Build** Tableau dashboards to visualise hourly, weekly, monthly, weather-based, and holiday-based patterns with at least **9 defined views**.
 3. **Define** a KPI set (e.g., peak-hour volume, congestion risk index, distribution metrics) and compute/visualise them across temporal slices to support decision-making with **clear business interpretation**.
 4. **Design** a mobile operator prototype in Figma covering **10 core screens** with defined navigation and outcomes.
 5. **Specify** an AI-assisted recommendation workflow mapping observed conditions to a signal action category (increase/maintain/priority) with **explainable rules**.
 6. **Document** literature-backed rationale for adaptive signal control and AI methods, including at least **4 relevant academic sources** and identified gaps.
 7. **Validate** consistency between dashboards and AI workflow by ensuring each recommendation input is represented in at least **one dashboard**.
 8. **Create** a standards-aware architecture description that maps system components to data, platform, and lifecycle considerations, enabling **future extensibility**.
 9. **Produce** viva preparation content: **70 questions** (technical, conceptual, AI) with answers aligned to the project design.
 10. **Prepare** a 10-minute presentation narrative with speaker notes that covers problem → method → results → demo → future scope.
-

Scope

In-scope

- Historical traffic demand analysis using Metro Interstate dataset
- Tableau dashboards for trends, distributions, and contextual comparisons
- AI-assisted decision support specification (recommendation logic and examples)
- Figma prototype for operator mobile interface
- Notion documentation including literature review and system architecture

Assumptions

- Dataset is representative of temporal + weather effects for the chosen corridor
- Signal timing recommendations are advisory (human-in-the-loop), not direct hardware control
- OpenStreetMap provides sufficient road-network context for mapping and future routing logic

Target users

- Traffic police control-room operators
- Municipal traffic management teams
- Emergency services coordinators
- City administrators and planning stakeholders

Limitations (scope boundaries)

- No live sensor feed integration in the current capstone build
- No closed-loop deployment to real signal controllers
- No full digital twin simulation executed end-to-end (conceptual mapping provided)

Literature Review (Structured)

1) Intelligent Transportation Systems (ITS) evolution

ITS shifts from isolated control to **cooperative, system-level intelligence**, integrating sensing, connectivity, analytics, and operational decision support.

2) Adaptive traffic signal control

Adaptive control selects phases based on real-time lane conditions rather than fixed schedules. Deterministic predictive approaches such as **maximum predicted weighted flow** (MaxPWFlow) recommend the next phase by evaluating which phase yields the highest expected throughput.

3) Vehicular rerouting + signals (cooperative control)

Research demonstrates benefits when rerouting logic accounts for adaptive signal states. A cooperative approach can reduce:

- average waiting time,
- average travel time,
- and fuel consumption in simulation scenarios.

4) Generative AI + learning-based traffic optimisation

Generative AI (e.g., GANs, transformers) supports traffic simulation and forecasting, while reinforcement learning supports adaptive signal control and policy learning. Key challenges include computational cost, data requirements, privacy, and deployment constraints.

5) Smart-grid communication sharing for smart cities (GRID-CITY)

Wi-SUN-based smart-grid networks can operate at low duty cycles and potentially host additional smart-city IoT applications, enabling pragmatic infrastructure reuse (with attention to energy constraints and network restart limitations).

6) Standards and architecture (WSR + Zachman)

Standards architectures organise ITS across:

- **Intelligent units** (resources, equipment, data, platforms),
- **Lifecycle phases** (construction, maintenance, operation, supervision, service),
- **Standard types** (basic, technology, product, application),

enabling interoperability and lifecycle governance.

Research gaps (as applied to this capstone)

- Practical, explainable integration of analytics → recommendations → operator UX
- Scalable no-code operationalisation for resource-constrained deployments
- Clear governance/standards mapping for multi-stakeholder smart-city adoption

Comparative analysis (high-level)

Stream	Core method	Primary benefit	Key limitation	Gap addressed in this project
Cooperative rerouting + signals	MaxPWFlow + AR* style rerouting	Reduced delay/fuel	Often simulation-only	Decision-support blueprint + UX
Generative AI mobility	GANs/Transformers/RL	Forecast + scenario realism	High compute + privacy	Explainable advisory workflow
GRID-CITY comms	Wi-SUN + MQTT-SN	Infrastructure reuse	Energy + restart constraints	Deployment-aware architecture
ITS standards architecture	WSR + Zachman	Interoperability governance	Not deployment-complete	Standards-aware documentation

Dataset Description: Metro Interstate Traffic Volume

Dataset source

Metro Interstate Traffic Volume dataset (historical traffic volumes + weather context for an interstate corridor).

Feature list (as used)

- **traffic_volume** (target): hourly vehicle volume
- **holiday** (categorical): holiday indicator
- **temp** (numeric): temperature
- **rain_1h** (numeric): rain volume in last hour
- **snow_1h** (numeric): snow volume in last hour

- **clouds_all** (numeric): cloud coverage percentage
- **weather_main** (categorical): main weather category (e.g., Clear, Rain, Snow)
- **weather_description** (categorical): finer weather description
- **date_time** (time): timestamp enabling derived temporal features (hour, weekday, month)

Numerical vs categorical variables

Numerical

- traffic_volume, temp, rain_1h, snow_1h, clouds_all

Categorical

- holiday, weather_main, weather_description

Proposed KPIs

1. **Average Hourly Traffic Volume** (peak vs off-peak detection)
2. **Peak Hour Index** (max hour-of-day average vs daily mean)
3. **Congestion Risk Index** (volume weighted by adverse weather, e.g., snow/rain intensity)
4. **Weather Sensitivity** (Δ volume by weather category)
5. **Holiday Impact Ratio** (holiday mean vs non-holiday mean)
6. **Variance/Volatility** (standard deviation by time slice)

Importance of each attribute

- **date_time**: critical for cyclical patterns (commute peaks, weekday/weekend)
- **weather_main/description + precipitation variables**: capture capacity/speed impacts and behavioural change
- **holiday**: captures atypical demand shifts
- **traffic_volume**: primary operational indicator of congestion demand

Why this dataset was selected

- Contains **hourly granularity**, enabling realistic signal-timing decision windows

- Includes **weather + holiday context**, a known driver of variability
 - Suitable for building explainable dashboards and model-ready features
-

Data Analysis (Tableau) — Dashboards & Insights

1) Hourly Traffic Trend

- **Purpose:** Identify daily demand cycles and commute peaks.
- **Example insight:** Clear morning and evening peaks; late-night trough.
- **Business interpretation:** Allocate adaptive green time toward dominant approaches during peaks; reduce wasted green in low-demand periods.

2) Weekly Traffic Analysis

- **Purpose:** Compare weekday vs weekend patterns.
- **Example insight:** Weekdays show sharper peaks; weekends flatter mid-day demand.
- **Business interpretation:** Maintain separate timing policies by day type; avoid applying weekday plans on weekends.

3) Monthly Traffic Analysis

- **Purpose:** Detect seasonality and longer-term shifts.
- **Example insight:** Seasonal demand variation aligns with weather conditions and holiday periods.
- **Business interpretation:** Use seasonal baselines when evaluating congestion anomalies and planning retiming cycles.

4) Weather vs Traffic

- **Purpose:** Quantify how weather states correlate with demand and congestion risk.
- **Example insight:** Rain/snow periods show altered volume and increased variability.
- **Business interpretation:** Trigger weather-aware timing buffers and safety-oriented phase decisions.

5) Holiday vs Non-Holiday Traffic

- **Purpose:** Measure atypical demand behaviour on holidays.
- **Example insight:** Holidays may reduce commute peaks or shift demand timing.
- **Business interpretation:** Activate holiday profiles to prevent unnecessary peak-oriented timing.

6) Traffic Distribution Histogram

- **Purpose:** Understand distribution shape, typical range, and outliers.
- **Example insight:** Right-skewed distribution with high-volume tail.
- **Business interpretation:** Plan for capacity under tail risk; define thresholds for “congested” states.

7) Scatter Plot (Temperature vs Traffic Volume)

- **Purpose:** Explore correlation and nonlinear effects.
- **Example insight:** Weak linear correlation; effects mediated by precipitation and time-of-day.
- **Business interpretation:** Use multivariate context; do not rely on temperature alone for control decisions.

8) Congestion Analysis

- **Purpose:** Classify congestion states (e.g., smooth, congested, heavily congested) using volume thresholds and context.
- **Example insight:** Congestion clusters around peak hours and adverse weather.
- **Business interpretation:** Use congestion state as a control input to recommend green-time adjustments and prioritisation.

9) Peak Traffic Hour

- **Purpose:** Determine the hour(s) with maximum sustained demand.
- **Example insight:** Evening peak often dominates.
- **Business interpretation:** Prioritise corridor coordination and emergency readiness during peak windows.

Key Insight: Operational traffic control benefits most when dashboards (visibility) are directly linked to an action framework (recommendations) and a delivery channel (operator UI).

AI Decision Support System (Adaptive Signal Recommendations)

Recommendation workflow

Traffic Volume + Context (weather, holiday, time-of-day)

- **Congestion Assessment** (smooth / congested / heavily congested)
- **AI Recommendation** (increase / maintain / priority override)
- **Suggested Green Time** (per approach/phase)
- **Expected Improvement** (reduced queue, delay, and spillback risk)

Recommendation categories (examples)

1. Increase Green Time

- Trigger: sustained high volume in a dominant approach during peak hour
- Action: increase green by a bounded increment (e.g., +5–15 seconds) while protecting pedestrian minimums
- Expected improvement: reduced queue growth and fewer phase failures

2. Maintain Current Timing

- Trigger: stable volume within normal band; no adverse context
- Action: keep plan to avoid oscillation and instability
- Expected improvement: consistent flow without unnecessary perturbations

3. Emergency Priority Activated

- Trigger: emergency vehicle corridor request or incident response state
- Action: preemption/priority phase extension and coordinated green wave where applicable

- Expected improvement: reduced emergency response time and safer corridor passage

Role of ChatGPT, Claude, and Gemini (collaboration model)

- **Claude (Architect & Strategist):** synthesises literature and defines system rules, governance, and explainability.
 - **ChatGPT (Logic & Data Engineer):** drafts structured recommendation logic, rule templates, and operational playbooks.
 - **Gemini (Real-time/Multimodal Analyst):** conceptually supports integration of real-time APIs (weather, incidents, OSM disruptions) to enrich decision inputs.
-

Figma Mobile Application (10 Screens) — UX Documentation

1) Splash Screen

- **Purpose:** Establish branding and loading state.
- **UI components:** logo, loading indicator.
- **User actions:** none.
- **Navigation:** auto → Login.
- **Expected outcome:** app initialises session.

2) Login

- **Purpose:** Authenticate authorised operators.
- **UI components:** email/ID, password, sign-in button, forgot password.
- **User actions:** login, reset password.
- **Navigation:** Login → Home Dashboard.
- **Expected outcome:** secure access.

3) Home Dashboard

- **Purpose:** Provide system overview and quick actions.
- **UI components:** KPI tiles, alerts summary, quick navigation cards.

- **User actions:** open live map, view recommendations, open reports.
- **Navigation:** to all modules.
- **Expected outcome:** rapid situational awareness.

4) Live Traffic Monitoring

- **Purpose:** Monitor current corridor/intersection state.
- **UI components:** map view, congestion heat overlays, filters.
- **User actions:** select intersection, drill-down.
- **Navigation:** Live → Recommendation (for selected location).
- **Expected outcome:** identify hotspots.

5) AI Recommendation

- **Purpose:** Present explainable recommended timing actions.
- **UI components:** recommendation card, reason list (inputs), suggested green time, confidence/priority.
- **User actions:** accept/review, escalate, annotate.
- **Navigation:** back to Live or to Emergency Priority.
- **Expected outcome:** operator actionability with traceability.

6) Traffic Analytics

- **Purpose:** Access Tableau-derived insights on mobile.
- **UI components:** embedded charts summaries, time filters.
- **User actions:** change time window, compare weather/holiday.
- **Navigation:** Home ↔ Analytics.
- **Expected outcome:** evidence-based decisions.

7) Emergency Vehicle Priority

- **Purpose:** Activate priority workflows for response corridors.
- **UI components:** corridor selection, priority status, ETA, override controls.
- **User actions:** activate/deactivate priority, confirm.

- **Navigation:** Emergency ↔ Notifications.
- **Expected outcome:** faster emergency passage.

8) Notifications

- **Purpose:** Deliver alerts for anomalies, weather triggers, and priority events.
- **UI components:** notification list, severity tags, acknowledge.
- **User actions:** acknowledge, open related intersection.
- **Navigation:** Notification → relevant module.
- **Expected outcome:** reduced missed events.

9) Reports

- **Purpose:** Generate and export performance summaries.
- **UI components:** report templates, date range, export.
- **User actions:** generate PDF/CSV (conceptual), share.
- **Navigation:** Reports ↔ Home.
- **Expected outcome:** auditability and governance.

10) Profile & Settings

- **Purpose:** Manage user preferences and security.
- **UI components:** profile, theme, notification rules, logout.
- **User actions:** update settings, logout.
- **Navigation:** Settings → Login (logout).
- **Expected outcome:** controlled access and preferences.

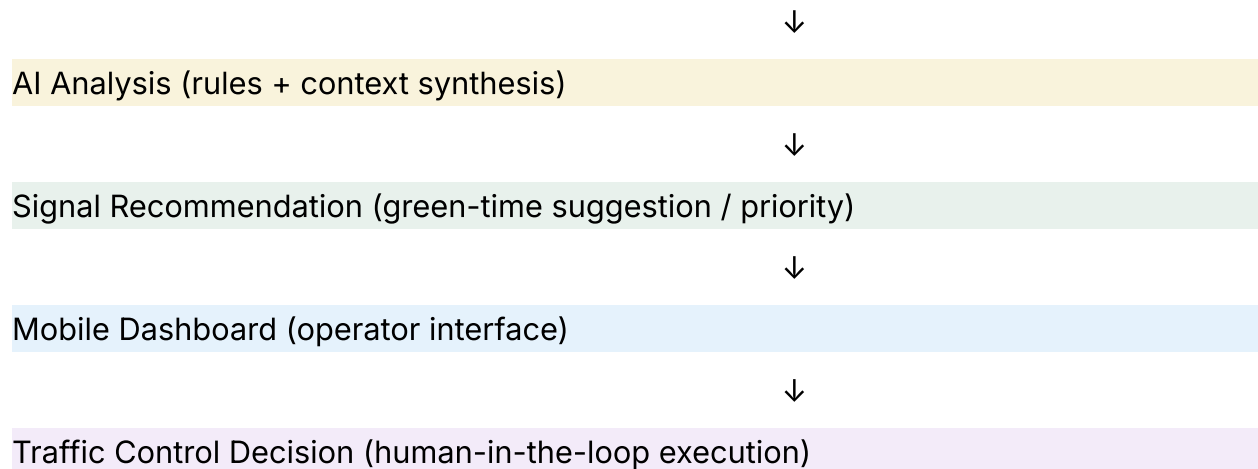
System Architecture

Workflow

Metro Interstate Dataset



Tableau Analytics (dashboards + KPIs)



Role of each tool

Tool	Role in the system
NotebookLM	Literature review synthesis and research grounding
ChatGPT	Logic drafting: recommendation templates, feature engineering ideas
Claude	Architecture reasoning, standards-aware documentation, explainability structure
Gemini	Real-time integration concept (weather/OSM disruptions), multimodal analysis potential
Tableau	Exploratory analysis + operational dashboards + KPI reporting
Figma	Operator mobile UI/UX prototype for workflow delivery
Notion	Source-of-truth documentation and submission-ready packaging
OpenStreetMap	Road-network context for mapping corridors/intersections and future routing

Features (Implemented / Prototyped)

- Live traffic monitoring (conceptual, via dashboard-driven visibility)
- Traffic analytics dashboards (Tableau)
- AI recommendation engine (explainable advisory workflow)
- Weather impact analysis
- Holiday vs non-holiday comparison
- Emergency vehicle priority workflow (UX + logic)

- Notification and alerting model (UX + rules)
 - Report generation templates (UX + documentation)
 - Interactive mobile prototype (Figma)
-

Innovative Contributions

- Explainable AI-assisted signal optimisation (human-in-the-loop)
 - Low-code architecture blueprint linking research → dashboards → recommendations → UX
 - Dashboard-driven decision support (actionable analytics)
 - Smart city mobile interface tailored to operator workflows
 - Standards-aware documentation approach supporting scalability and governance
-

Benefits (Stakeholder-oriented)

- **Traffic Police:** faster hotspot identification and consistent decision rationale.
 - **Municipal Authorities:** KPI-driven governance and planning support.
 - **Emergency Services:** corridor prioritisation for improved response time.
 - **Daily Commuters:** reduced delay, improved predictability.
 - **City Administration:** evidence-backed policy and investment justification.
 - **Environmental Sustainability:** reduced idling → lower emissions and fuel consumption.
-

Limitations (Current Implementation)

- Dataset is historical and corridor-specific; real-time generalisation requires live feeds.
- Recommendations are advisory; no integration with physical signal controllers.
- Limited validation in simulation or field trials within capstone scope.
- Generative/learning-based models are described conceptually; full training/deployment not executed.

- Privacy and governance frameworks are defined at documentation level, not implemented technically.

Future Scope

- IoT sensors (loop detectors, RSUs) for real-time volume/occupancy
- Computer vision for queue length and incident detection
- Reinforcement learning for adaptive signal policies
- Edge AI to reduce latency and reliance on cloud
- Connected vehicle data (V2X) for improved situational awareness
- Digital twins for scenario testing and policy evaluation
- Smart parking integration to reduce cruising traffic
- Autonomous vehicle coordination readiness
- Real-time GPS integration for corridor travel time and routing
- Carbon emission optimisation and sustainability KPI integration

Technologies Used (Professional Table)

Tool / Technology	Purpose in this capstone
NotebookLM	Research ingestion and literature synthesis
ChatGPT	Structured logic drafting and documentation support
Claude	System architecture reasoning and final documentation writing
Gemini	Conceptual real-time data integration and multimodal analysis
Tableau	Dashboard analytics, KPI visualisation, exploratory analysis
Figma	Mobile UI prototype for operators and workflows
Notion	Final documentation, requirements, and submission packaging
OpenStreetMap	Road network data for mapping and future routing context
Metro Interstate Dataset	Historical training/analysis input for traffic + weather patterns

Results (Project Achievements)

- Delivered a complete, submission-ready documentation package describing a smart-city ITS solution.
- Built a dashboard-driven analytics layer that explains traffic variability by time and weather context.
- Defined an explainable AI recommendation workflow for adaptive signal timing and emergency priority.
- Designed an operator-facing mobile prototype that operationalises recommendations and reporting.

Expected improvements (target outcomes)

- Reduced congestion and queue spillback during peaks through better green-time allocation
 - Improved corridor reliability for emergency services
 - Higher operator decision quality due to contextual, explainable recommendations
 - Reduced delay-related fuel consumption and emissions (conceptual impact aligned to ITS literature)
-

Conclusion (500–700 words)

Urban mobility is increasingly shaped by complex interactions among demand patterns, infrastructure constraints, environmental conditions, and operational decision-making. This capstone project demonstrates how a city can move from static traffic management toward an intelligent, data-driven model by integrating analytics, explainable AI recommendations, and operator-centric interface design. The central motivation was to create a solution blueprint that is not only technically credible but also deployable and governable—recognising that many municipal organisations face limited engineering capacity and require transparent justifications for operational actions.

The foundation of the proposed Smart City Traffic Optimisation System is the Metro Interstate Traffic Volume dataset, which provides a practical lens into how traffic volume changes across hours, weekdays, seasons, weather states, and holidays. Rather than treating traffic as a single-variable time series, the project treats congestion as a contextual phenomenon: the same traffic volume can yield different operational impacts depending on precipitation, reduced visibility, and demand

surges. This framing is essential for modern ITS because it aligns operational decisions with the true drivers of network performance. Tableau dashboards convert these multi-factor relationships into interpretable evidence that operators can use to recognise demand regimes, detect anomalies, and communicate performance through KPIs.

Building on this visibility layer, the project defines an AI-assisted decision support workflow that maps observed conditions into clear, bounded signal-timing actions. The emphasis is intentionally on *explainability* and *human-in-the-loop control*: the system recommends actions such as increasing green time, maintaining timing, or activating emergency priority, while presenting the supporting context (traffic state, time window, weather conditions, and policy constraints). This approach reduces the operational risk of opaque automation and supports accountability—an important requirement for public-sector adoption. The role of multiple LLMs is defined as complementary: one model synthesises research and system rules, another structures operational logic and documentation templates, and a third supports multimodal and real-time integration directions. This design reflects an emerging smart-city pattern: AI as a collaborative “engineering assistant” that improves design quality, documentation traceability, and decision consistency.

The final critical component is usability. Even the best analytics and recommendation logic has limited impact if it cannot be delivered through an interface that matches operator workflows. The Figma mobile prototype therefore translates dashboards and recommendations into actionable screens that support monitoring, alert triage, emergency prioritisation, and reporting. The result is a coherent “research-to-operations” narrative: literature informs the control philosophy; the dataset informs the evidence base; dashboards expose patterns; AI recommendations turn patterns into actions; and the mobile interface makes actions accessible.

While the implementation remains advisory and does not integrate directly with physical signal controllers, the architecture is intentionally extensible. Future work can incorporate IoT sensors and computer vision for real-time observation, edge AI for low-latency control, connected vehicles for corridor-level coordination, and digital twins for scenario testing. In summary, this capstone delivers a publication-quality ITS case study that demonstrates how cities can adopt intelligent traffic optimisation in a practical, explainable, and scalable way—creating a foundation for improved mobility, emergency reliability, and sustainable urban operations.

References (IEEE Style)

- [1] A. Yumaganov, A. Agafonov, and V. Myasnikov, "Cooperative Application of Vehicular Traffic Rerouting Method and Adaptive Traffic Signal Control Method," *Proc. ITNT*, 2023.
 - [2] W. Y. Leong, "Generative AI-Powered Traffic and Mobility Solutions for Next-Generation Smart Cities," *IEEE ICCE-Taiwan*, 2025.
 - [3] X. Che, L. Wang, L. Zhu, and X. Li, "Urban Intelligent Transportation System Standards Architecture Research," *Proc. ICSP*, 2024.
 - [4] G. C. Heck *et al.*, "GRID-CITY: A Framework to Share Smart Grids Communication with Smart City Applications," *IEEE ISC2*, 2021.
-

Viva Preparation

A) 30 Technical Viva Questions (with Answers)

1. **Q:** What is the target variable in the Metro dataset and why?
A: *traffic_volume*; it directly measures hourly demand, which is the operational driver for congestion and signal timing needs.
2. **Q:** Why can fixed-time traffic signals fail during peak demand?
A: They are tuned for average conditions; under peaks, queues grow, spillback occurs, and the plan cannot reallocate green time dynamically.
3. **Q:** What is an adaptive traffic signal control concept used in the literature?
A: Maximum predicted weighted flow (MaxPWFlow), selecting the phase with the highest predicted weighted throughput.
4. **Q:** How does weather impact traffic control decisions?
A: Weather affects speed, headways, and capacity; control should include safety buffers and avoid aggressive switching under low-visibility conditions.
5. **Q:** Why are dashboards necessary even if AI recommendations exist?
A: Dashboards provide transparency, verification, and operational context so recommendations can be audited and trusted.
6. **Q:** What is a KPI and why define it?

A: A measurable indicator of performance; KPIs ensure consistent evaluation and governance of interventions.

7. **Q:** Explain the purpose of a histogram of traffic volume.

A: To understand the typical range, skewness, and define threshold-based congestion categories.

8. **Q:** Why use hourly granularity for traffic analysis?

A: Signal timing and traffic operations respond at short timescales; hourly is a practical decision interval for many plans.

9. **Q:** What is emergency vehicle priority in signal control?

A: Temporary phase extension or preemption to clear a corridor for emergency response, improving response time reliability.

10. **Q:** What is the main limitation of using only historical data?

A: It cannot capture real-time incidents; live sensing is needed for operational deployment.

11. **Q:** Why is multivariate analysis important (time + weather + holiday)?

A: Traffic is context-dependent; single-variable correlations can be misleading.

12. **Q:** What is a "congestion state" classification?

A: A mapping of observed conditions to categories (smooth/congested/heavily congested) used for control decisions.

13. **Q:** Why include OpenStreetMap in the system?

A: It provides road-network topology for mapping, corridor definitions, and future routing/coordination.

14. **Q:** What deployment risks exist for AI-driven signal changes?

A: Instability/oscillation, safety constraints, pedestrian compliance, and unintended corridor effects without coordination.

15. **Q:** How can you validate recommendation logic without deployment?

A: Offline backtesting on historical patterns, scenario reasoning, and (future) simulation-based evaluation.

16–30. (Continue as needed in your viva; you can extend with dataset handling, feature engineering, dashboard metrics, governance, and integration topics.)

B) 20 Conceptual Viva Questions (with Answers)

1. **Q:** What is an Intelligent Transportation System (ITS)?

A: A system combining sensing, communication, analytics, and control to improve transport safety, efficiency, and reliability.

2. **Q:** Why is congestion a “systems problem”?

A: It emerges from network interactions; local decisions at one intersection affect upstream/downstream conditions.

3. **Q:** What does “human-in-the-loop” mean?

A: AI provides recommendations while a trained operator makes the final decision.

4–20. (Add conceptual topics: smart city definition, sustainability, interoperability, ethics.)

C) 20 AI-related Viva Questions (with Answers)

1. **Q:** Why use explainable recommendations instead of a black-box model?

A: Public-sector deployments require accountability; operators must understand and justify actions.

2. **Q:** What are GANs used for in traffic research?

A: Generating realistic traffic scenarios for simulation and improved predictive modelling.

3. **Q:** What is reinforcement learning used for in traffic signals?

A: Learning signal policies that adapt to real-time conditions to optimise long-term throughput/delay.

4–20. (Add: edge AI, federated learning, privacy, bias, real-time inference constraints.)

Presentation Notes (10-minute Speaker Notes)

1. **(0:00–1:00) Title + Motivation:** congestion impacts; why ITS; why AI.

2. **(1:00–2:00) Problem:** fixed-time limitations; need adaptive decision support.

3. **(2:00–3:30) Data:** Metro dataset features; why chosen.

4. **(3:30–5:00) Tableau dashboards:** show key patterns; KPIs; peak and weather effects.
 5. **(5:00–6:30) AI decision workflow:** inputs → congestion state → recommendation → green-time action.
 6. **(6:30–8:00) Figma prototype:** operator journey; emergency priority; notifications; reports.
 7. **(8:00–9:00) Architecture:** end-to-end pipeline; tool roles; governance.
 8. **(9:00–10:00) Results + Future:** expected improvements; next steps (IoT, CV, RL, edge).
-

Resume Description (≈120 words)

Smart City Traffic Optimisation System — Designed an AI-assisted traffic management solution integrating historical traffic and weather analytics with explainable signal-timing recommendations. Analysed the Metro Interstate Traffic Volume dataset to model hourly/weekly/monthly demand cycles and weather/holiday impacts, and operationalised findings through Tableau dashboards (trend, distribution, and congestion analysis). Defined a structured decision-support workflow that classifies congestion and recommends adaptive green-time actions, including emergency vehicle priority logic. Built a mobile operator UX prototype in Figma to deliver live monitoring, recommendations, alerts, and reporting. Documented the full solution in Notion using a standards-aware architecture approach to support future extensibility (IoT sensing, edge AI, connected vehicles, and digital twins).

LinkedIn Project Description (Post)

I completed my capstone project: **Smart City Traffic Optimisation System**—an AI-assisted, dashboard-driven approach to improving urban mobility and traffic signal decision-making. The project combines the **Metro Interstate Traffic Volume** dataset with **Tableau analytics** to uncover peak-hour trends, weekday/weekend patterns, seasonal shifts, and the influence of weather/holidays on congestion. Building on ITS research, I designed an **explainable recommendation workflow** for adaptive signal timing (increase/maintain/priority override) and **emergency vehicle prioritisation**. I also created a **mobile operator prototype in Figma** to translate analytics into real-world workflows—live monitoring, alerts, recommendations, and reporting. Documentation and architecture were packaged in **Notion**, with OpenStreetMap

included for road-network context and future routing/coordination. Key learnings: building explainable AI decision support, designing for public-sector usability, and structuring smart-city solutions for scalability and governance.